

# Charan Mogulla

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## Summary

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AI Engineer specializing in production-grade LLM applications, RAG architectures, and scalable AI systems. Experienced in designing multi-agent workflows, fine-tuning large language models (LoRA/QLoRA), and deploying containerized AI services on Azure with robust MLOps practices. Proven track record of improving response accuracy by 40% and reducing enterprise document review time by 50% through end-to-end AI solution delivery.

## Professional Experience

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### Staffingtree Inc

**Aug 2023 — Present**

*AI Engineer*

*Overland Park, KS*

- Architected and productionized enterprise-grade LLM applications using LangChain and OpenAI, serving real-time customer workflows with scalable API infrastructure and robust monitoring.
- Designed advanced RAG pipelines with vector databases and embedding optimization, improving contextual response accuracy by 40% while reducing hallucinations.
- Built multi-agent orchestration systems to automate complex customer operations, translating business logic into reliable AI-driven workflows.
- Implemented LoRA and QLoRA fine-tuning pipelines for domain adaptation, improving task-specific performance while optimizing GPU utilization and inference cost.
- Established evaluation and feedback loops (automated testing, prompt benchmarking, human-in-the-loop validation) to continuously improve response quality. Containerized AI services with Docker and deployed via CI/CD pipelines to ensure reproducible, scalable, secure production rollouts.

### Infosys

**Jul 2022 — Jul 2023**

*AI Engineer*

*Hyderabad, India*

- Architected and deployed LLM-powered conversational AI systems on Microsoft Azure, leveraging Azure OpenAI Service, Azure Machine Learning, and Azure App Services, enabling intelligent automation for enterprise customer engagement platforms serving 100K+ users.
- Designed and implemented RAG-based enterprise knowledge assistants using LangChain and Azure Cognitive Search (vector search), improving contextual accuracy by 35% and significantly reducing hallucinations in production.
- Built scalable AI pipelines using Azure ML, handling data ingestion, embedding generation, model experimentation, evaluation tracking, and secure model deployment.
- Integrated Azure OpenAI + LangChain orchestration workflows to automate call center query resolution, reducing average resolution time by 30% and lowering operational costs.
- Built and deployed containerized AI microservices using FastAPI and Docker on Azure Kubernetes Service (AKS), implementing Azure DevOps CI/CD, model versioning, monitoring, prompt evaluation frameworks, and automated retraining to ensure scalable, secure, and production-grade AI systems.

### HCL Technologies

**Dec 2020 — Jun 2022**

*Data Engineer*

*Hyderabad, India*

- Architected and maintained scalable data pipelines processing large-scale enterprise logistics and transaction datasets, enabling downstream analytics and AI model consumption.
- Designed distributed real-time streaming systems using Apache Kafka and Spark Streaming for fraud detection, improving detection accuracy and reducing response latency by 25%.
- Built robust ETL workflows for structured and unstructured data, implementing data validation, schema enforcement, and performance optimization to ensure high data reliability.
- Developed LLM-driven document summarization pipelines for enterprise logistics reports, reducing manual review time by 50% and accelerating operational decision-making.

- Automated large-scale data preprocessing and feature engineering pipelines for ML models, containerized workflows with Docker, and partnered with ML/DevOps teams to productionize scalable, monitored, and governance-compliant data systems — improving inference speed by 25% and ensuring high availability.

## Technical Skills

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- **AI & LLM Systems:** LangChain, Azure OpenAI, RAG, Vector Databases (FAISS, Qdrant), LoRA, QLoRA, Prompt Engineering, Multi-Agent Systems, Anthropic Claude API, Langfuse
- **Machine Learning:** PyTorch, Deep Learning, NLP, Computer Vision, Model Evaluation, Fine-Tuning, HuggingFace Transformers
- **Data & Distributed Systems:** Apache Kafka, Spark Streaming, ETL Pipelines, SQL, Large-Scale Data Processing, Databricks
- **Cloud & DevOps:** Microsoft Azure, AWS, Azure ML, AKS, Docker, Kubernetes, CI/CD, MLOps, Monitoring & Observability, Fly.io, Terraform
- **Programming:** Python, C++, FastAPI, Flask, TypeScript, Next.js

## AI & LLM Projects

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### Checkmate — AI Code Review GitHub App [Live](#) · [checkmate-ai.fly.dev](#) · [repo](#)

- Built a production GitHub App that reviews pull requests with Claude Sonnet 4.6, validates inline diff comments, and uses repo-aware RAG (Qdrant + BGE embeddings) for context.
- Deployed FastAPI + RQ worker stack on Fly.io with managed Neon Postgres, Upstash Redis, and Qdrant Cloud — clickable demo URL, \$0/mo on free tiers.
- Instrumented end-to-end LLM observability with Langfuse (traces, cost, latency) and a 14-case *promptfoo* evaluation suite wired into CI.

### AI-Powered Chatbot with LangChain & RAG

- Built custom RAG pipelines using FAISS & OpenAI GPT-4 for context-aware chatbot retrieval.
- Integrated LangChain agents for document search & summarization in customer service chatbots.
- Developed a FastAPI-based API for chatbot deployment with real-time response generation.

### LLM Fine-Tuning for Domain-Specific Chatbot

- Fine-tuned Mistral & LLaMA models using LoRA/QLoRA for enterprise AI assistants.
- Optimized prompt engineering techniques, improving response relevance by 35%.

### AI-Powered Search Engine with LLM Summarization

- Implemented search augmentation with GPT-4 & T5, providing concise AI-generated summaries.
- Integrated LangChain retrievers & embeddings for improved search relevance.

## Education

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### University of Central Missouri

*Master of Science, Computer Science in AI & ML*

**Aug 2023 — May 2025**

*Lee's Summit, Missouri*

- **GPA:** 3.6/4.0
- **Coursework:** AI Ethics, Machine Learning, NLP, Deep Learning, Data Analytics

### Bharat Institute of Engineering and Technology

*Bachelor of Science, Computer Science, Engineering*

**2017 — 2021**

*Hyderabad, India*

- **GPA:** 8.3/10
- **Coursework:** AI, Computer Vision, Networking, Machine Learning

## Certifications

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- LangChain for Developers — LangChain Official (2024)
- Generative AI with Large Language Models — Coursera (2024)
- Deep Learning Specialization — Andrew Ng (2023)